

ELECTRONICS FINAL NOTE

BY AMZAD



PDF By Musa Khan

① Definition & types of FET:

The field-effect transistor is a type of transistor that uses an electric field to control the flow of current in a semiconductor. ~~device~~

It comes two types:

① Junction FET (JFET).

② Metal oxide semiconductor FET (MOSFET).

② JFET definition & its construction:

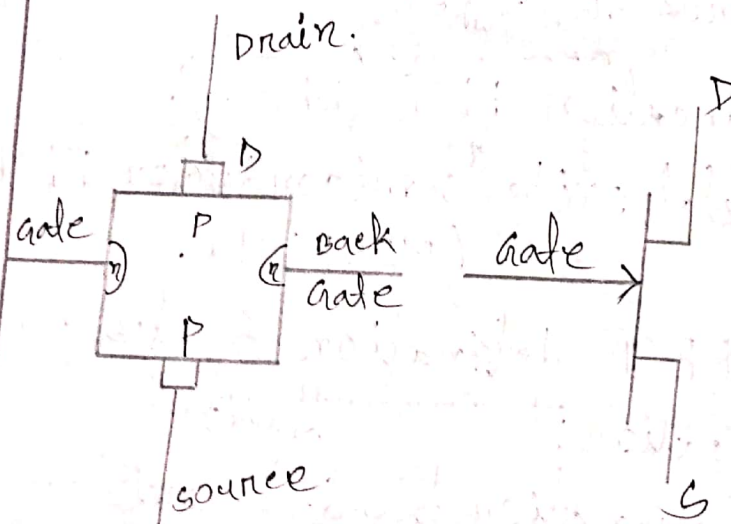
The junction field-effect transistor is one of the simplest types of field-effect transistor. JFETs are ~~are~~ three-terminal semiconductor ~~or~~ devices that can be used as

electronically controlled switches
or resistors, or to build amplifiers.

JFET $\Rightarrow$ $\begin{cases} \text{P-JFET} \\ \text{n-JFET} \end{cases}$

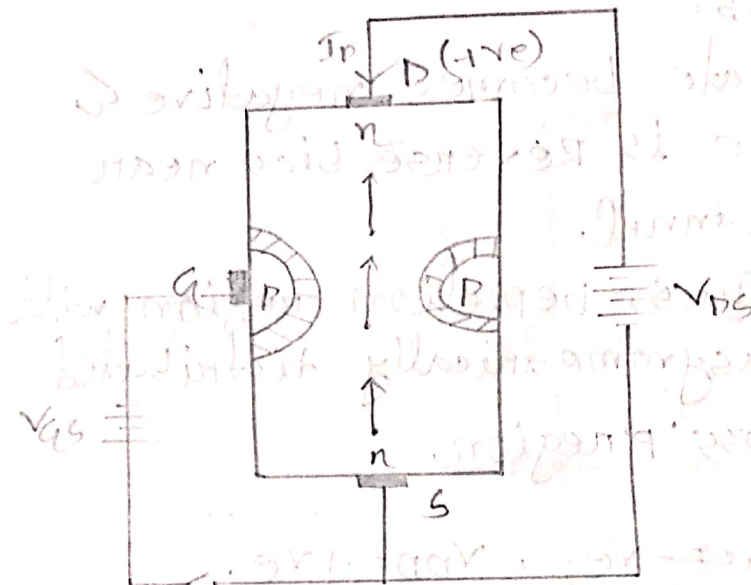
3 terminal device,

① Drain ② Source ③ Gate



n-JFET.

③ Working principle of nJFET:



case: n-channel JFET.

(i) $V_{GS} = 0, V_{DS} = 0$

$\rightarrow$ Depletion region are of equal thickness and it is symmetrically distributed.

ii) $V_{GS} = 0$, $V_{DD} = +ve$.
→ There is voltage drop from D to S.

→ Gate becomes negative & there is reverse bias near P-channel.

→ Thus, depletion region will be asymmetrically distributed across 'P' region.

iii) $V_{GS} = -ve$, $V_{DD} = +ve$.

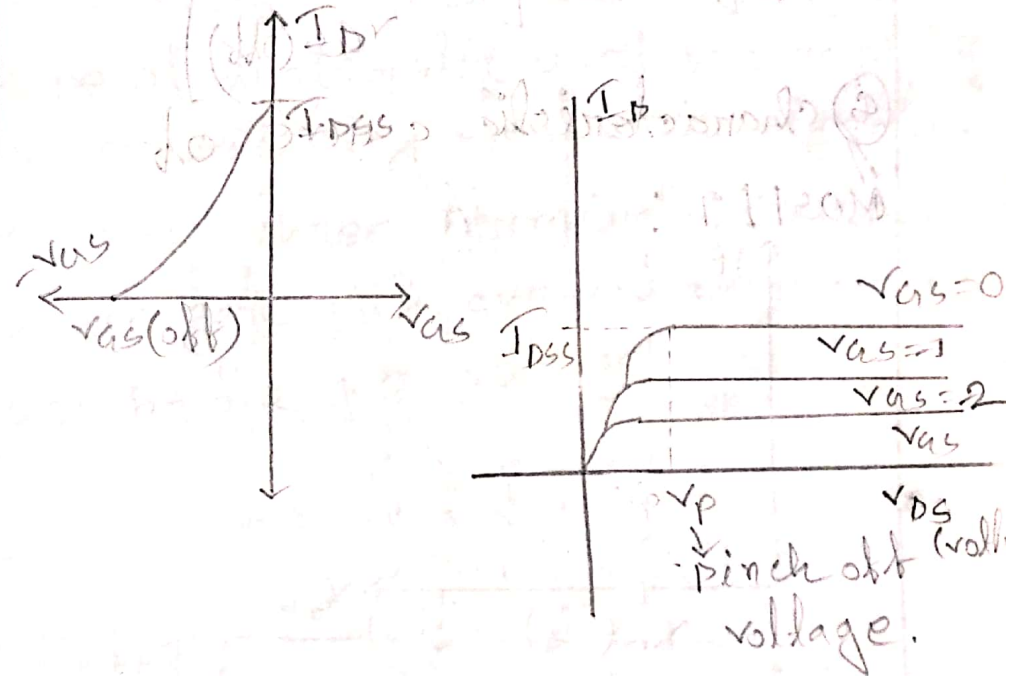
→ electron does not flow from G to D.

$$\therefore I_D = 0$$

$$\rightarrow V_{GS} = V_P$$

= Pinch off voltage.

④ characteristics curve of JFET
(i/p & o/p chara...):



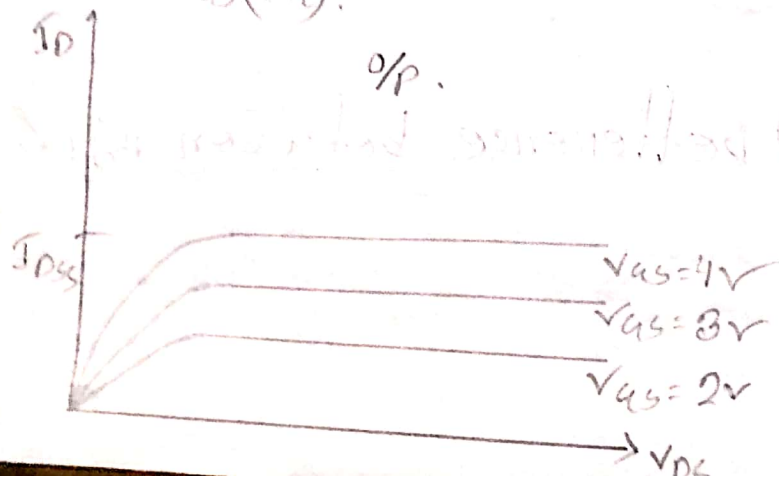
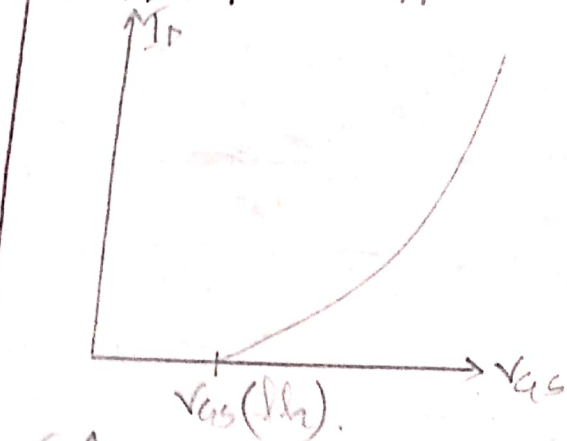
⑤ Difference between BJT & FET.

JFET MATH

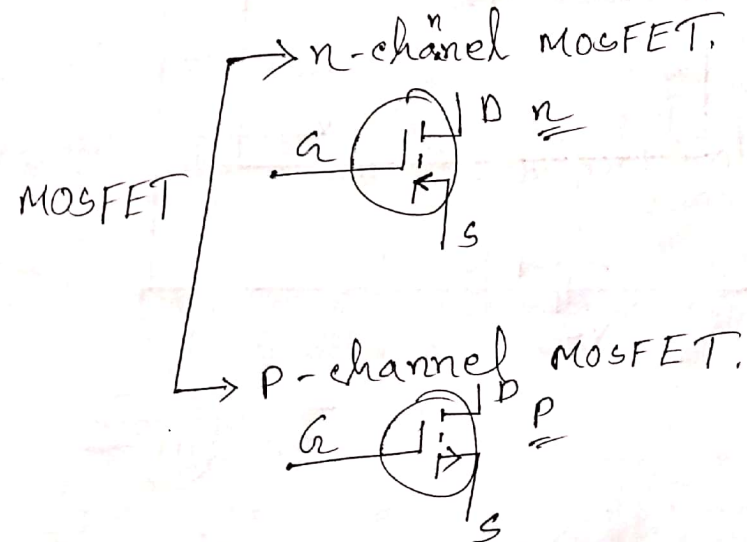
$$① V_p = |V_{as(0bb)}|$$

$$② I_D = I_{DSS} \left[1 - \frac{V_{as}}{V_{as(0bb)}} \right]^2$$

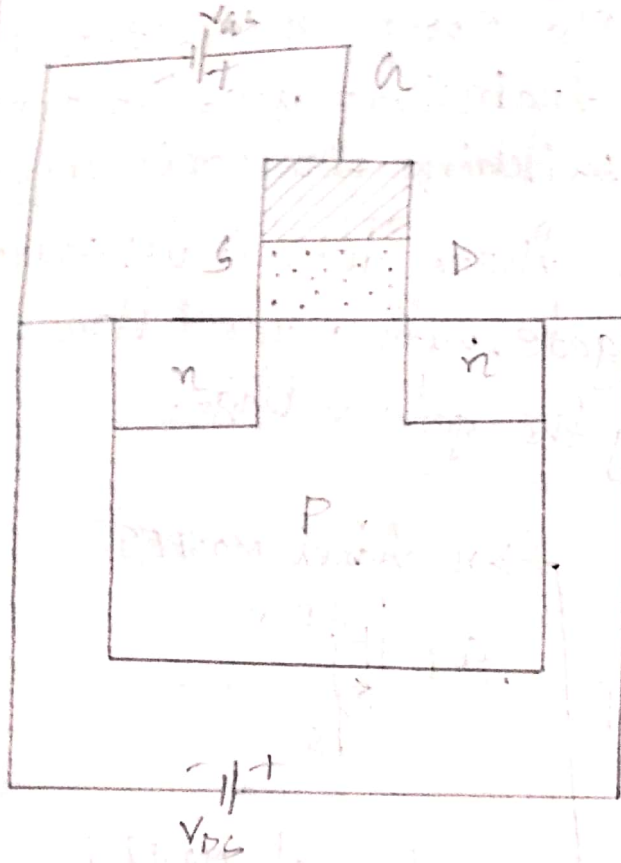
③ characteristic curve of MOSFET:



④ MOSFET definition & its type:
A MOSFET (Metal-Oxide-Semiconductor field-effect transistor) is a type of transistor used for amplifying or switching electronic signals. It has three terminals: source, drain, gate, with current flow controlled by the gate voltage.



⑦ Enhancement MOSFET construction & working principle:



MATH of 25.

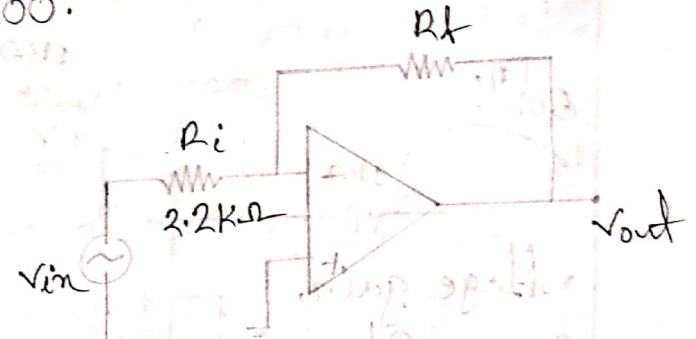
25.25: Given the op-amp configuration in the fig, determine the value of R_f required to produce a closed-loop voltage gain of -100.

Soln:

$$A_{CL} = -\frac{R_f}{R_i}$$

$$\text{or, } -100 = -\frac{R_f}{2.2}$$

$$\therefore R_f = 220 \text{ k}\Omega$$



25.26: Determine the output voltage.

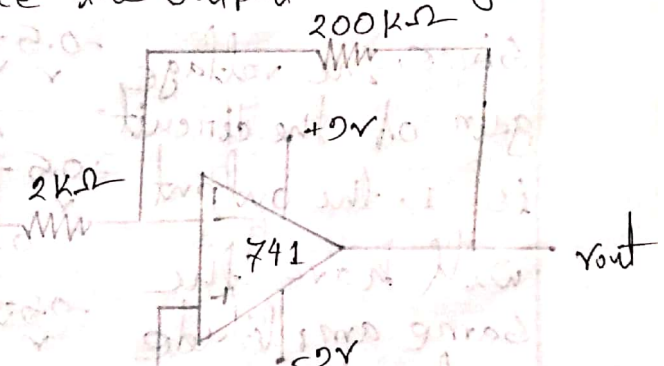
Soln:

$$A_{CL} = -\frac{R_f}{R_i}$$

$$= -\frac{200}{2}$$

$$= -100$$

$\therefore$ output voltage: 2.5 mV
 $V_{out} = A_{CL} \times V_{in}$



$$= (100) \times (2.5 \text{ mV})$$

$$= -250 \text{ mV}$$

$$= -0.25 \text{ V}$$

25.27: Find output voltage.

Soln:

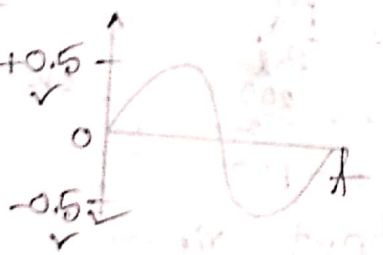
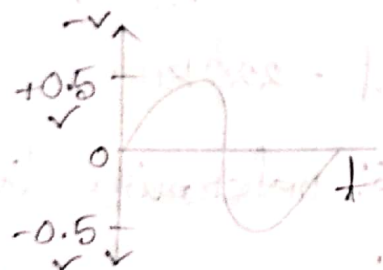
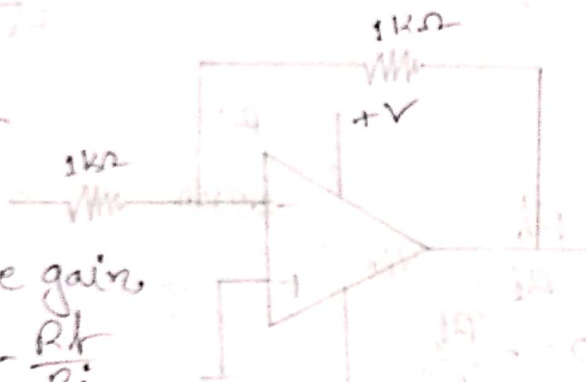
voltage gain

$$A_{CL} = -\frac{R_f}{R_i}$$

$$= -\frac{1}{1}$$

$$\therefore A_{CL} = -1$$

Since, the voltage gain of the circuit is -1, the output will have the same amplitude but with 180° ,



25.28: Find the output voltage.

Soln:

voltage

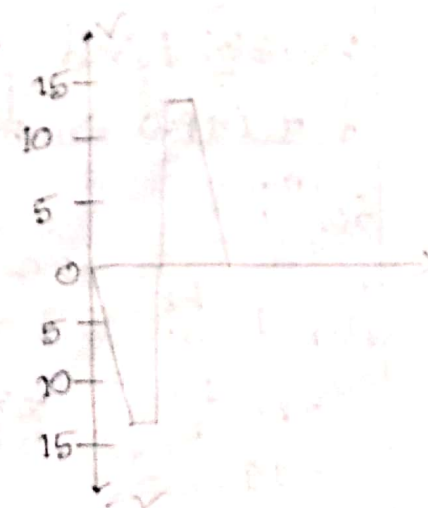
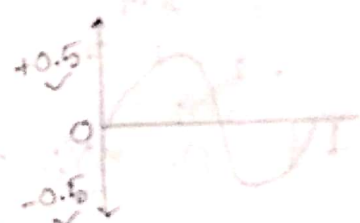
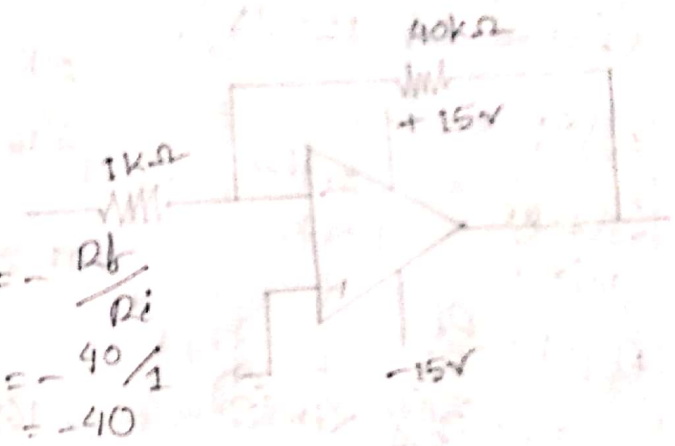
$$\text{gain, } A_{CL} = -\frac{R_f}{R_i}$$

$$= -\frac{40}{1}$$

$$= -40$$

$$\therefore \text{Max o/p} = 15 - 2$$

$$= 13(1)$$



25.32: Calculate the output voltage.
(input of 120mV).

Solⁿ:

Voltage gain,

$$A_{CL} = 1 + \frac{R_f}{R_i}$$

$$= 1 + \frac{240}{2.4}$$

$$= 1 + 100$$

$$= 101$$

$$V_{out} = A_{CL} \times V_{in} = (101) \times (120\text{mV})$$

$$= 12.12\text{mV}$$

25.33: Find output voltage.

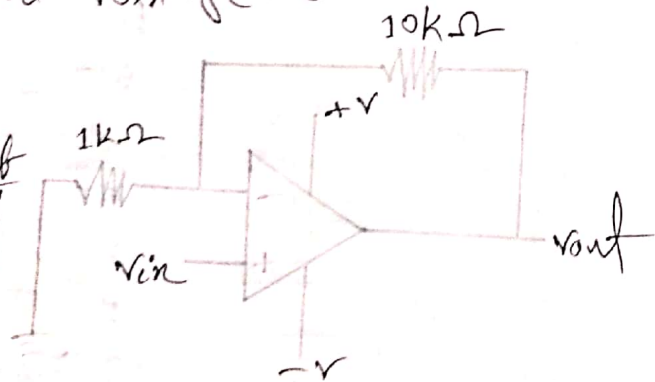
for input voltage. (i) 1V & (ii) -1V.

Solⁿ:

$$A_{CL} = 1 + \frac{R_f}{R_i}$$

$$= 1 + \frac{10}{1}$$

$$= 11$$



(i) For $V_{in} = 1\text{V}$,

$$V_{out} = A_{CL} \times V_{in}$$

$$= 11 \times 1$$

$$= 11\text{V}$$

(ii) For $V_{in} = -1\text{V}$,

$$V_{out} = A_{CL} \times V_{in}$$

$$= 11 \times -1$$

$$= -11\text{V}$$

25.34: For non inverting amplifier circuit show in the fig. find peak-to-peak output voltage.

Solⁿ:

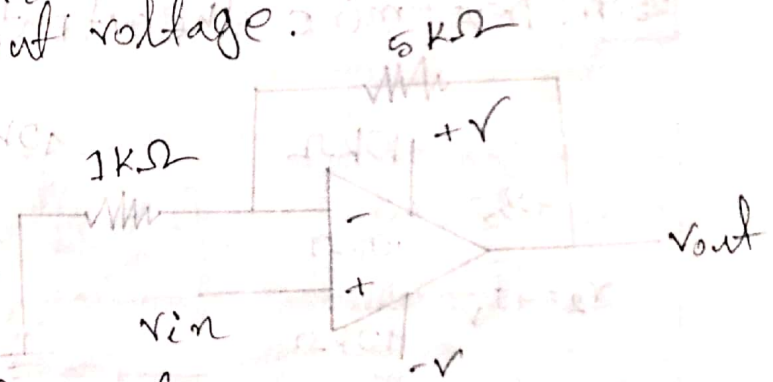
The input signal is

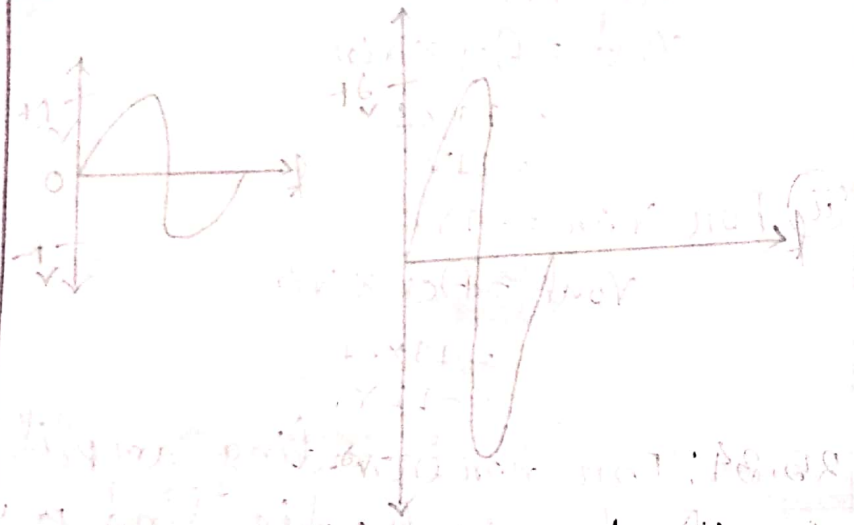
2V peak-to-peak.

Voltage gain, $A_{CL} = 1 + \frac{R_f}{R_i} = 1 + \frac{5}{1} = 6$.

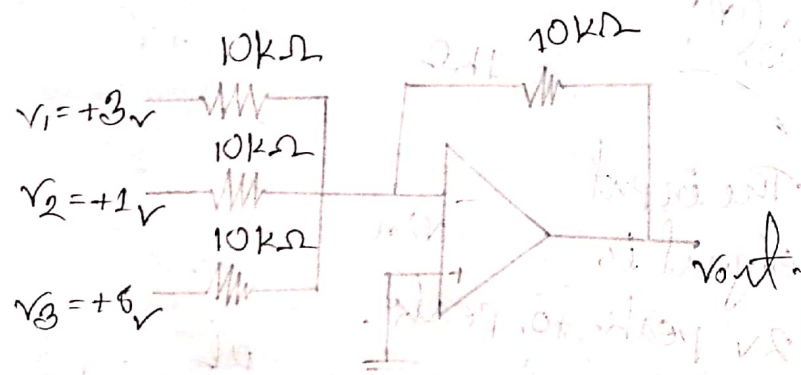
∴ peak-to-peak output voltage.

$$= 6 \times 2 = 12\text{V}$$



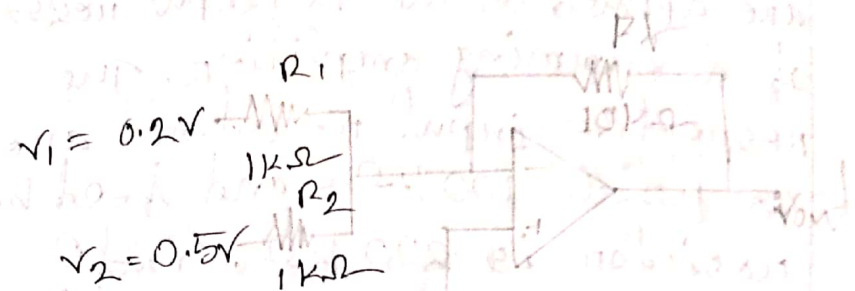


25.44: Determine the output voltage.



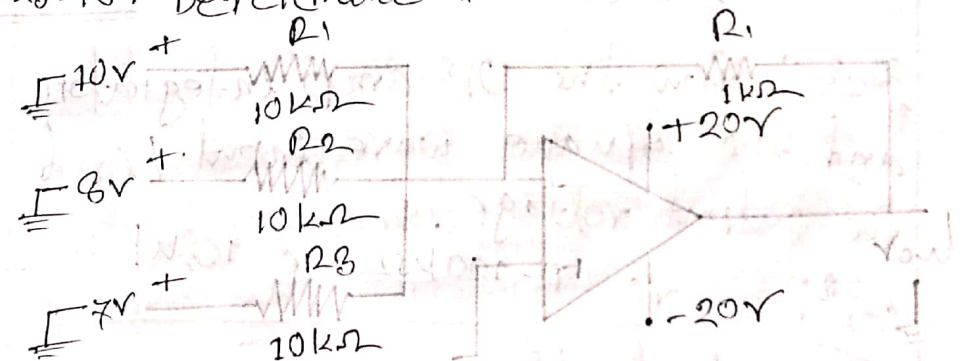
$$\begin{aligned} \therefore V_{out} &= -(V_1 + V_2 + V_3) \\ &= -(3 + 1 + 6) \\ &= -12V. \end{aligned}$$

25.45: Determine the output voltage.



$$\begin{aligned} \therefore V_{out} &= -\frac{R_f}{R_1} (V_1 + V_2) \\ &= -\frac{10}{1} (0.2 + 0.5) \\ &= -7V. \end{aligned}$$

25.46: Determine the output voltage.



$$\begin{aligned} \therefore V_{out} &= -\frac{R_f}{R} (V_1 + V_2 + V_3) \\ &= -\frac{1}{10} (10 + 8 + 7) = -2.5V. \end{aligned}$$

25.47: Two voltage of $+0.6 \text{ V}$ & -1.4 V are applied to the two input resistors of a summing amplifier. The respective input resistors are $400 \text{ k}\Omega$ & $100 \text{ k}\Omega$ and feedback resistor is $200 \text{ k}\Omega$. V_{out} ?

Solⁿ:

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

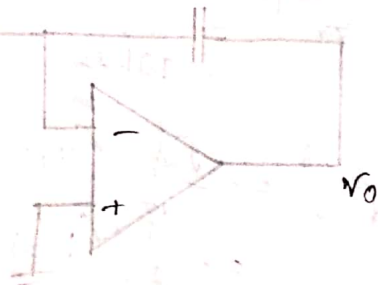
$$= -200 \left(\frac{0.6}{400} + \frac{-1.4}{100} \right)$$

$\therefore V_{out} = 2.5 \text{ V}$

25.50: Show the OP-amp integrator and the square wave input. Find the output voltage.

Solⁿ: V_i $R=100 \text{ k}\Omega$ $C=10 \mu\text{F}$

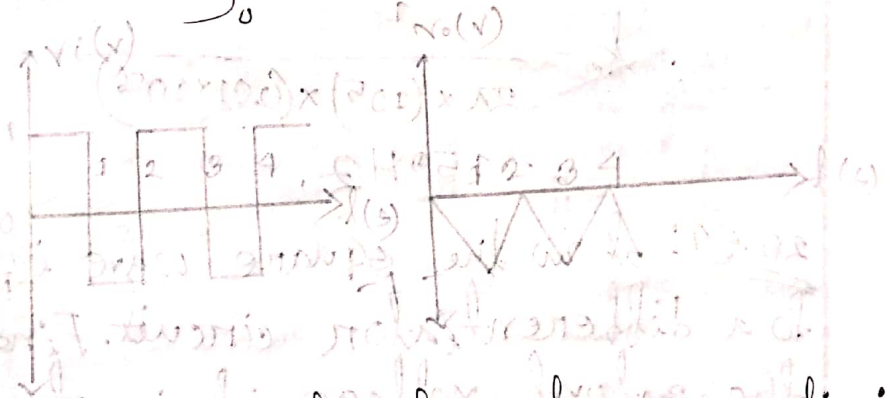
The output voltage of the circuit is given by-



$$V_o = -\frac{1}{R_C} \int_0^t V_i dt$$

$$R_C = (100 \text{ k}\Omega) \times (10 \mu\text{F}) = (100 \times 10^3) \times (10 \times 10^{-6} \text{ F})$$

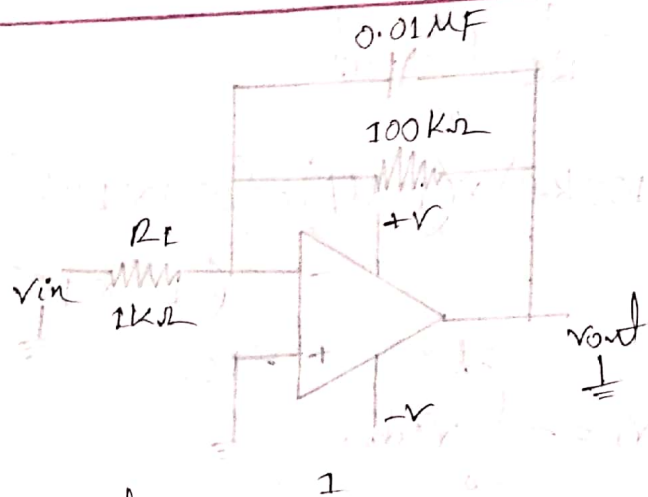
$$\therefore V_o = -\int_0^t V_i dt \quad = 15.$$



25.51: Determine the lower frequency limit (critical frequency) for the integrator circuit. show in the fig.

Solⁿ:

$$f_c = \frac{1}{2\pi R_f C}$$



$$\therefore f_c = \frac{1}{2\pi \times (10^5) \times (0.01 \times 10^{-6})}$$

$$= 159 \text{ Hz}$$

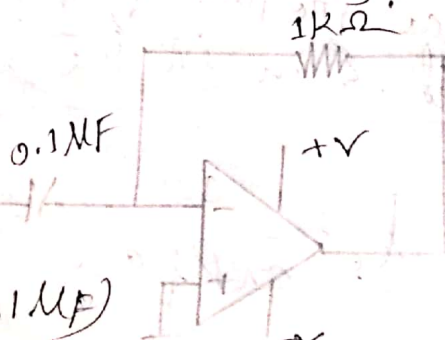
25.54: Show the square wave input to a differentiator circuit. Find the output voltage if input goes from 0V to 5V in 0.1ms.

Solⁿ:

output voltage 0.1μF

$$v_o = -R_c \frac{dv_i}{dt}$$

$$R_c = (1k\Omega) \times (0.1\mu F)$$



$$= (10^3) \times (0.1 \times 10^{-6})$$

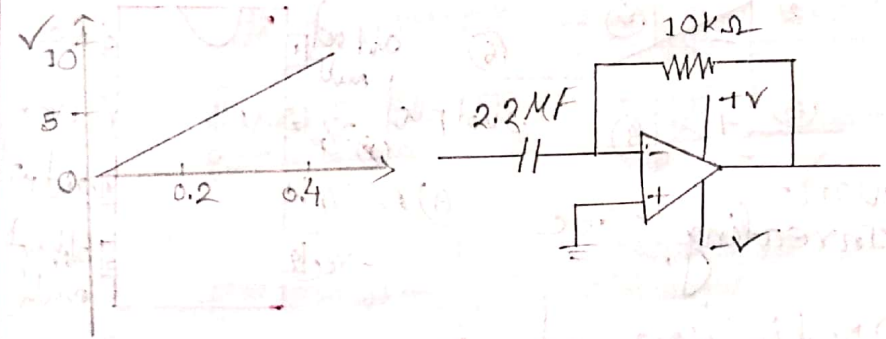
$$= 0.1 \times 10^{-3} \text{ s}$$

$$\text{Also, } \frac{dv_i}{dt} = \frac{5V}{0.1ms} = \frac{5 \times 10^4 V}{s} = 5 \times 10^4 \text{ V/s}$$

$$\therefore v_o = -(0.1 \times 10^{-3})(5 \times 10^4)$$

$$= -5V$$

25.55:



Solⁿ: output voltage, $v_o = -R_c \frac{dv_i}{dt}$

$$R_c = (10k\Omega) \times (2.2\mu F)$$

$$= (10^4 \Omega) \times (2.2 \times 10^{-6} F)$$

$$= 2.2 \times 10^{-2} \text{ s}$$

$$\frac{dv_i}{dt} = \frac{(10-0)V}{0.4s} = \frac{10V}{0.4s}$$

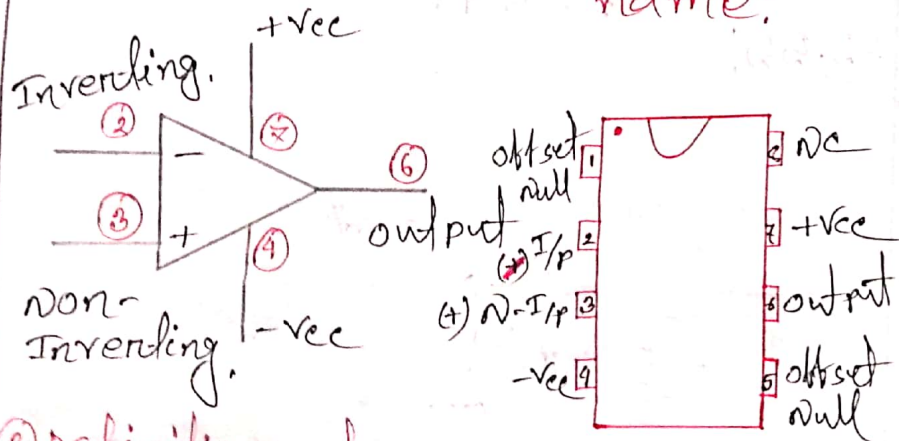
$$= 25 \text{ V/s}$$

$$V_o = -(2.2 \times 10^{-2}) \times 25$$

$$= -0.55 \text{ V}$$

The output voltage stays constant at -0.55 V .

① OP-Amp simple 6 terminal name:



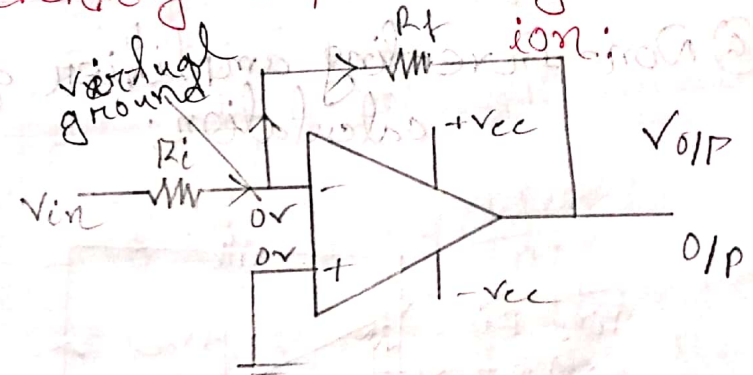
② Definition of OP-Amp:

An operational amplifier (OP-amp) is a high-gain electronic device to amplify voltage differences between its input terminals.

③ Ideal characteristics of OP-Amp.

- *1) Infinite open loop gain.
- *2) Infinite input impedance.
- *3) Zero output impedance.
- 4) Infinite bandwidth.
- 5) perfect balance.

⑤ Inverting amplifier gain calculation:



Let's,

$$I_{in} = \frac{V_{in} - 0}{R_i} = \frac{V_{in}}{R_i}$$

$$I_f = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f}$$

So,

$$I_{in} = I_f$$

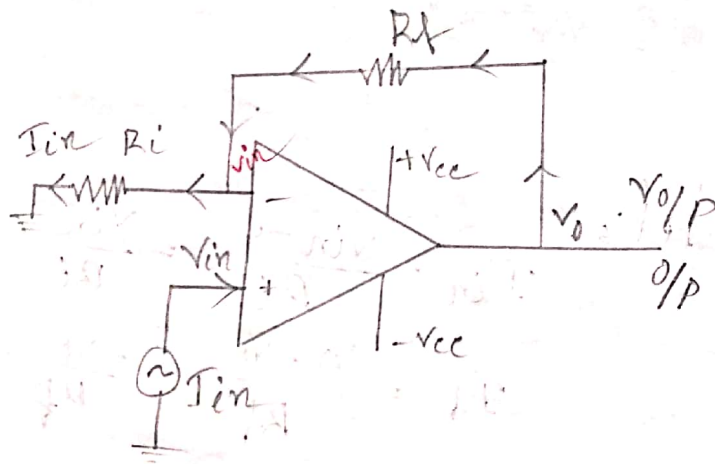
$$\text{or, } \frac{v_{in}}{R_i} = -\frac{v_o}{R_f}$$

$$\text{or, } \frac{R_f}{R_i} = -\frac{v_o}{v_{in}}$$

$$\text{or, } \frac{v_o}{v_{in}} = -\frac{R_f}{R_i}$$

$$\therefore A_{CL}(\text{inv.}) = -\frac{R_f}{R_i}$$

⑥ Non-Inverting amplifier gain calculation:



$$I_i = \frac{v_{in} - 0}{R_i} = \frac{v_{in}}{R_i}$$

$$I_f = \frac{v_o - v_{in}}{R_f} = \frac{v_o - v_{in}}{R_f}$$

We can write,

$$I_{in} = I_f$$

$$\text{or, } \frac{v_{in}}{R_i} = \frac{v_o - v_{in}}{R_f}$$

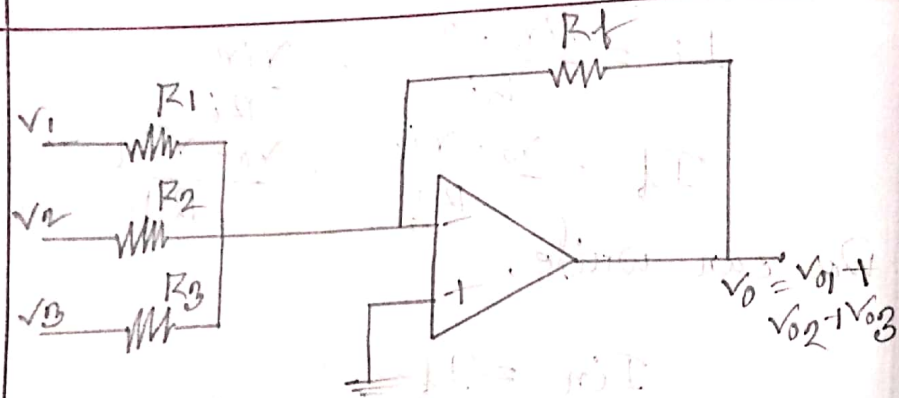
$$\text{or, } \frac{R_f}{R_i} = \frac{v_o - v_{in}}{v_{in}}$$

$$\text{or, } \frac{R_f}{R_i} = \frac{v_o}{v_{in}} - 1$$

$$\text{or, } \frac{v_o}{v_{in}} = 1 + \frac{R_f}{R_i}$$

$$\therefore A_{CL}(\text{non-inv.}) = 1 + \frac{R_f}{R_i}$$

⑦ Adding / Summing amplifier gain calculation:



$$V_{01} = -\frac{R_f}{R_1} \times V_1 \quad \left[\because I = \frac{V}{R} \right]$$

$$V_{02} = -\frac{R_f}{R_2} \times V_2$$

$$V_{03} = -\frac{R_f}{R_3} \times V_3$$

$$\therefore V_{01} + V_{02} + V_{03} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

$$\text{If, } R_1 = R_2 = R_3 = R,$$

$$\therefore V_0 = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

If in case,

$$R_f = R_1 = R_2 = R_3,$$

$$V_0 = -\frac{R_f}{R} (V_1 + V_2 + V_3),$$

$$\therefore V_{out} = -(V_1 + V_2 + V_3)$$

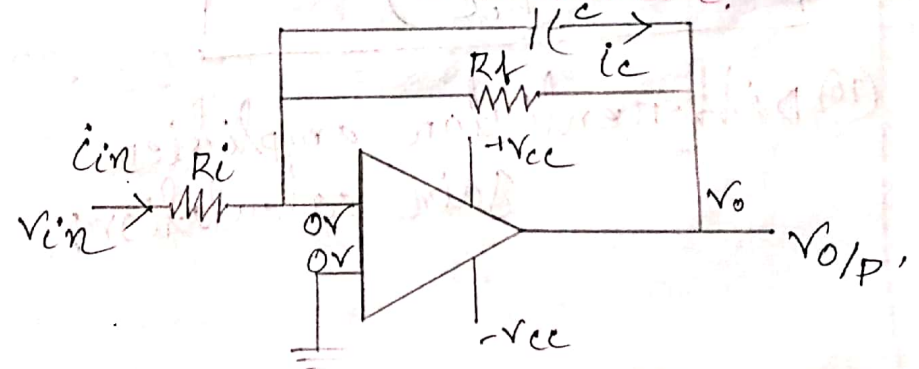
⑧ Average amplifier gain calculation:
From Q. After 7,

$$R_1 = R_2 = R_3 = 3R_f$$

$$\text{Or } V_0 = -\frac{R_f}{3R_f} (V_1 + V_2 + V_3)$$

$$\therefore V_0 = -(V_1 + V_2 + V_3)/3$$

⑨ Integrator amplifier gain calculation:



$$I_{in} = \frac{V_{in} - 0}{R_i} = \frac{V_{in}}{R_i}$$

$$i_c = C \frac{dv_c}{dt} = C \frac{d(-V_o)}{dt} = -C \frac{dV_o}{dt}$$

let,

$$I_{in} = i_c$$

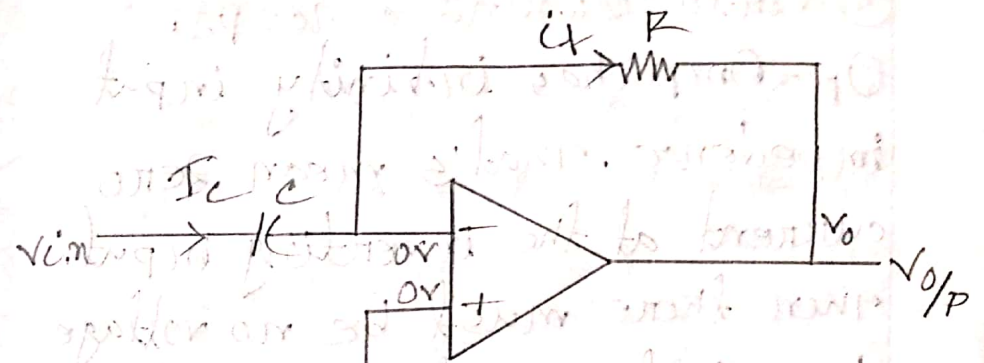
$$\text{or, } \frac{V_{in}}{R_i} = -C \frac{dV_o}{dt}$$

$$\text{or, } V_{in} = -R_i C \frac{dV_o}{dt}$$

$$\text{or, } \int V_{in} dt = -R_i C dV_o$$

$$\therefore V_o = -\frac{1}{R_i C} \int V_{in} dt$$

⑩ Differentiator amplifier gain calculation:



$$i_c = C \frac{dv_c}{dt}$$

$$= C \frac{dV_{in}}{dt}$$

$$i_f = \frac{0 - V_o}{R}$$

$$= -\frac{V_o}{R}$$

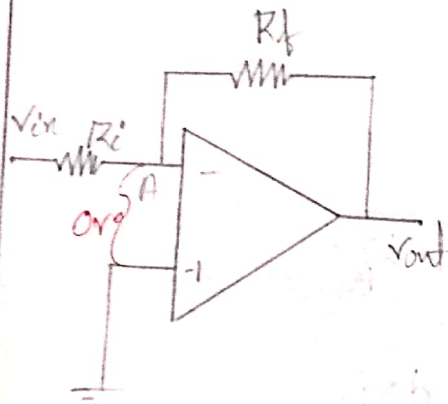
$$i_c = i_f$$

$$\text{or, } C \frac{dV_{in}}{dt} = -\frac{V_o}{R}$$

$$\therefore V_o = -R C \frac{dV_{in}}{dt}$$

④ Virtual Ground concept:

Op-Amp has infinity input impedance. That's mean zero current at the inverting input. Then there must be no voltage drop between the inverting and non-inverting inputs. The OV at the inverting input terminal is referred to as virtual ground.



ch - 19 .

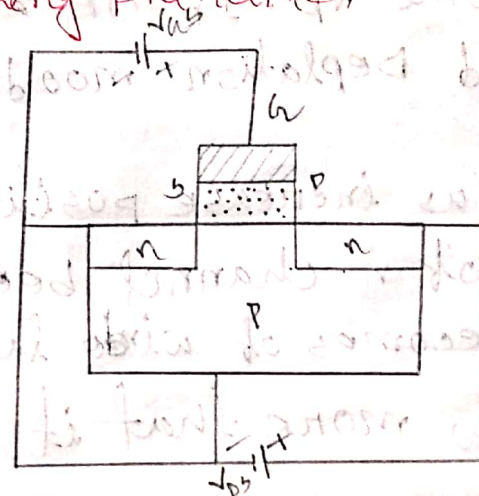
MATHS →
19.1, 2, 3, 5.

* Difference between BJT & FET,

soln:

<u>BJT</u>	<u>FET</u>
① i/p impedance low.	① - - - High.
② current control device.	② voltage - - - .
③ o/p impedance high.	③ - - - low.
④ switching speed low.	④ - - - High.

* Enhancement MOSFET construction & working principle.



Working Principle:

At $V_{GS} = 0V$ the current flow I_D become maximum and it called I_{DSS} current. When V_{GS} increase negatively, at the n conductor bar become negative carriers. because of positive charge in layer, layer become narrow and I_D decreasing as V_{GS} increase.

V_{GS} become V_p . $I_D = 0A$ and it called Depletion mood.

When, V_{GS} increase positively, the layer of n -channel become wide, becomes of wide layer I_D flow more than it is

more than I_{DSS} . And it called enhancement mode.

Problems

13.5 (u)

18.1 $\rightarrow$ 18.2

(knee concept)

⊗ 14.6

2M?

cha - 18.

MATH $\Rightarrow$ 18.1 - 18.4.

circuit which can turn ON or OFF current in an electrical circuit is known as a switching circuit.

① Definition of switching circuit:

A switching circuit is an electronic setup that controls the flow of electrical signals by opening or closing different paths in a circuit using electronic switches like transistors or relays.

② Types of switching circuit:

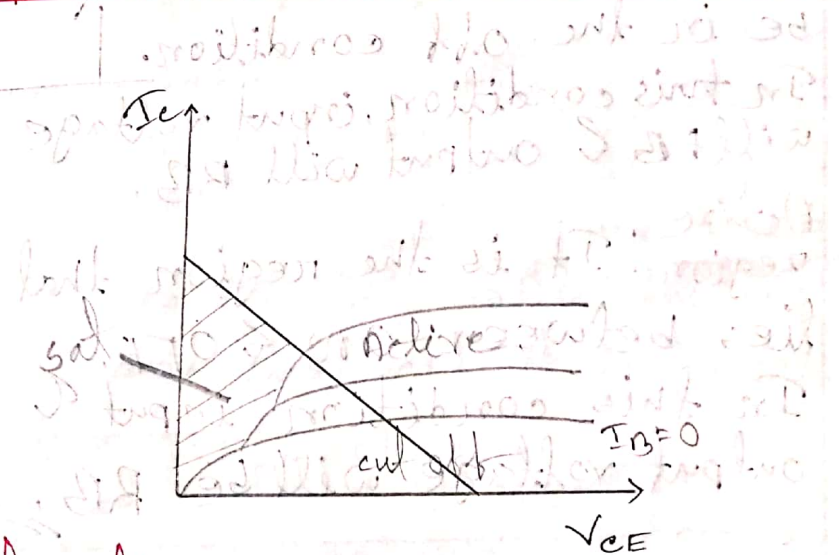
3 types:

- Mechanical switch.
- Electro-mechanical switch or Relay.
- Electronic switch.

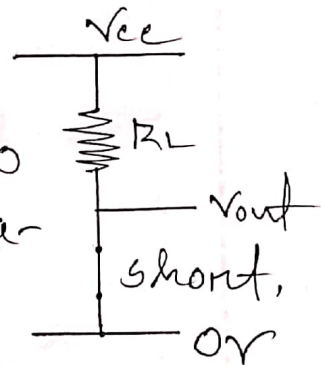
③ Switching action of a transistor.

off
ON
Active. } Region.

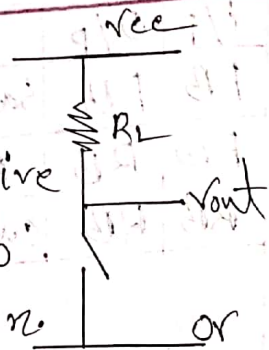
i/P	O/P	state
FB	FB	saturation $\rightarrow$ ON switch $\rightarrow$ short.
FB	RB	Active $\rightarrow$ Amplifier.
RB	RB	cut off $\rightarrow$ off switch $\rightarrow$ Open.



Saturation: (ON switch) When the input voltage is made so much positive that saturation collector current flows, the transistor is said to be in the ON condition. Both are in Forward bias.



cut off (off-switch): When the input voltage is zero or negative the transistor is said to be in the off condition. In this condition, input voltage will be 0 & output will be R_B .



Active region: It is the region that lies between ON & OFF. In this condition input & output voltage will be R_B .

Chap - 13.

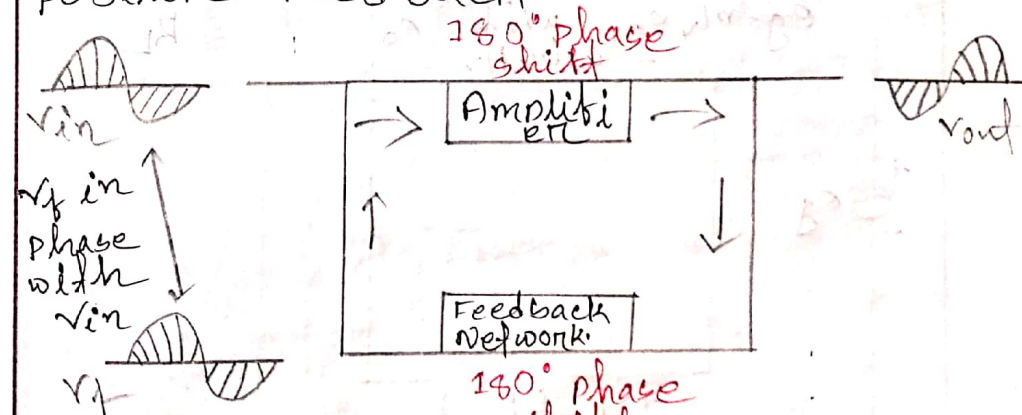
MATH $\Rightarrow$ 13.2 - 13.5.

* Feedback definition & its types:

The process of injecting a fraction of output energy of some device back to the input is known as feedback.

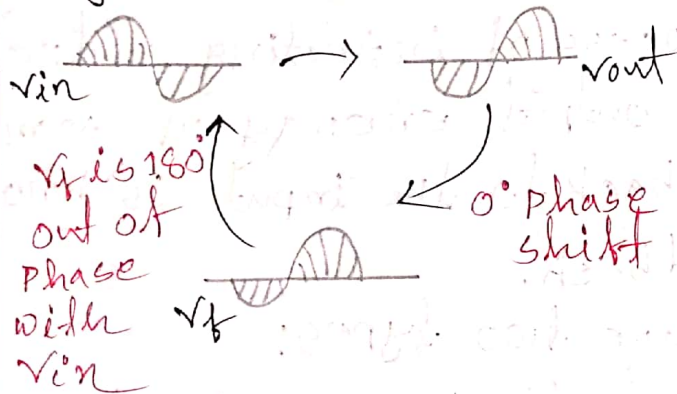
There are two types:

(i) **Positive feedback:** When the feedback energy is in phase with the input signal and thus aids it, it is called positive feedback.

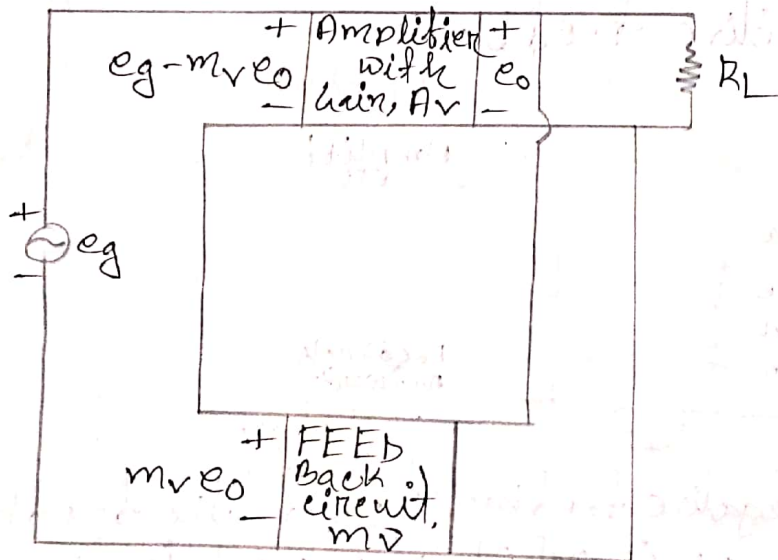


(ii) **Negative feedback:** When the feedback energy is out of phase with the input signal and thus opposes it, it is called negative feedback.

negative feedback.



* Gain of Negative voltage Feedback Amplifier:



Gain, A_v .

Output voltage, e_o .

Actual input to amplifier = $e_g - m_v e_o$.

So, $(e_g - m_v e_o) A_v = e_o$

or, $A_v e_g - A_v m_v e_o = e_o$

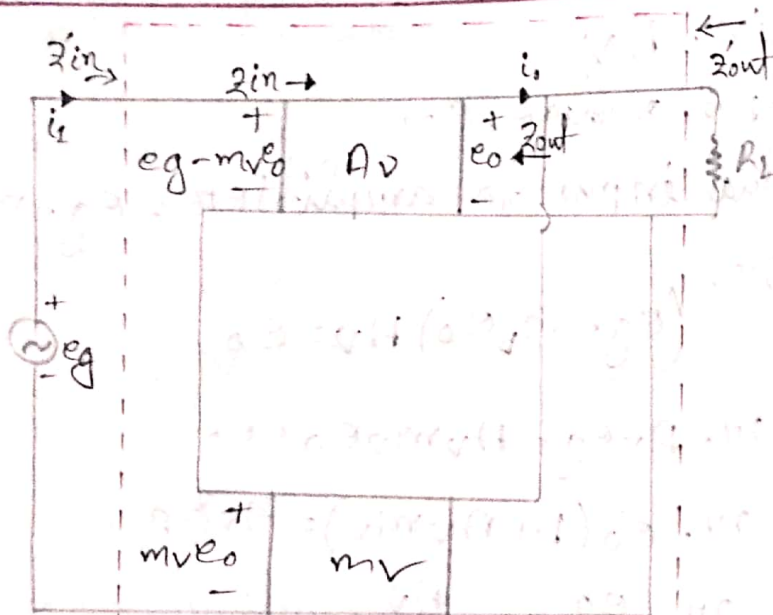
or, $e_o (1 + A_v m_v) = A_v e_g$

or, $\frac{e_o}{e_g} = \frac{A_v}{1 + A_v m_v}$

But, e_o/e_g is the voltage gain of the amplifier with feedback.

$$\therefore A_{vf} = \frac{A_v}{1 + A_v m_v}$$

* Show that, input impedance after negative voltage feedback increases...



$$e_g - m_v e_o = i_1 Z_{in}$$

$$e_g = (e_g - m_v e_o) + m_v e_o$$

$$= (e_g - m_v e_o) + A_v m_v (e_g - m_v e_o)$$

$$= (e_g - m_v e_o) (1 + A_v m_v) \quad \left[\because e_o = A_v (e_g - m_v e_o) \right]$$

$$= i_1 Z_{in} (1 + A_v m_v)$$

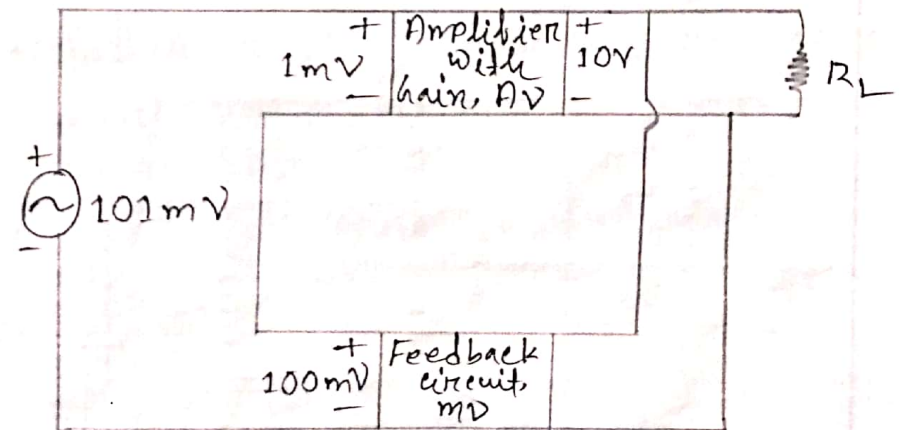
$$\left[\because e_g - m_v e_o \right] = i_1 Z_{in}$$

$$\text{or, } \frac{e_g}{i_1} = Z_{in} (1 + A_v m_v)$$

But, $e_g / i_1 = Z'_{in}$, the input impedance of the amplifier with negative voltage feedback.

$$\therefore Z'_{in} = Z_{in} (1 + A_v m_v)$$

⊕ Principle of Negative voltage Feedback in Amplifier:



The output of the amplifier is 10V.
 mV is 100 mV,
 1 mV appears at the input amplifier is 1 mV.

Gain of amplifier without feedback,

$$A_v = \frac{10V}{1mV} = 10,000.$$

Fraction of output voltage

$$\text{feedback, } m_v = \frac{100mV}{10V} \\ = 0.01.$$

Gain of amplifier with negative

$$\text{feedback, } A_{vf} = \frac{10V}{101mV} = 100 //$$

Chap - 14.

MATH $\rightarrow 14.5 / 14.6$

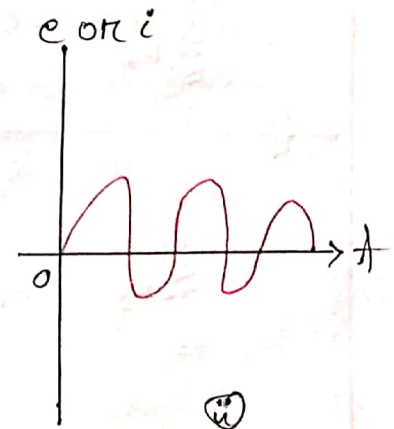
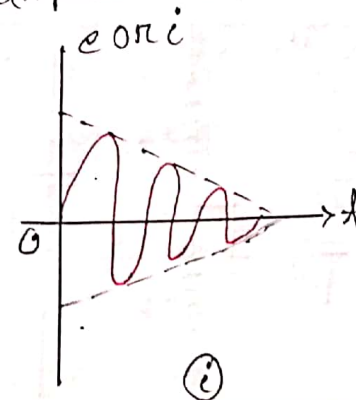
④ Sinusoidal Oscillator & its type:

An electronic device that generates sinusoidal oscillations of desired frequency is known as a sinusoidal oscillator.

2 types:

(i) Damped oscillations: The electrical oscillations whose amplitude goes on decreasing with time are called damped oscillation.

(ii) Undamped oscillations: The electrical oscillations whose amplitude remains constant with time are called undamped oscillation.

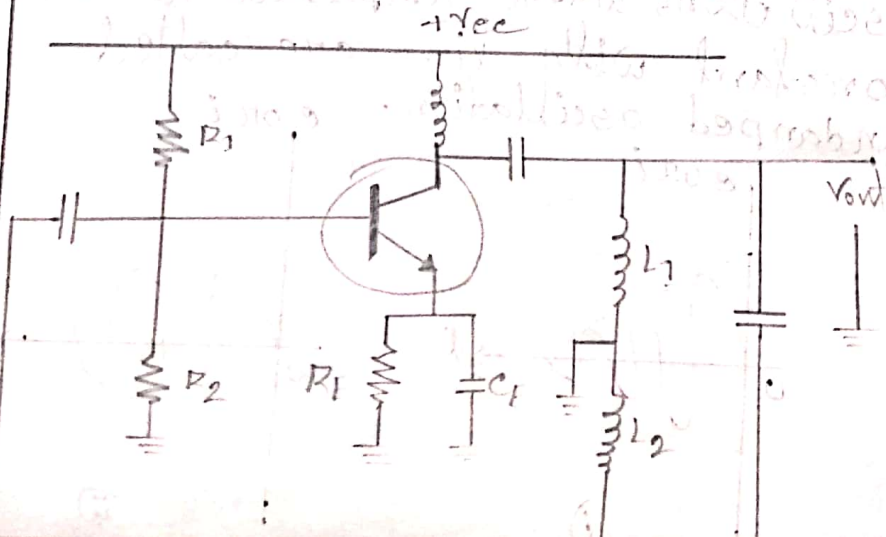


⊗ Different Types of Transistor Oscillators:

- | | |
|---------------------------------|------------------------------|
| (i) Tuned collector oscillator. | (ii) Colpitts oscillator. |
| (iii) Hartley oscillator. | (iv) Phase shift oscillator. |
| (v) Wien Bridge oscillator. | (vi) crystal oscillator. |

⊗ Hartley circuit details & operation:

The tank circuit is made up of L_1 , L_2 and C .



$$f = \frac{1}{2\pi\sqrt{C L_T}}$$

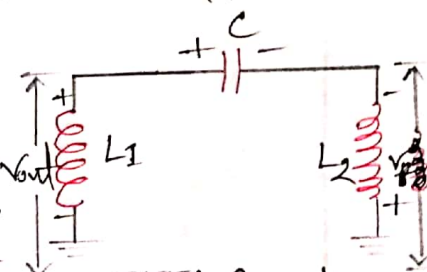
where,

$$L_T = L_1 + L_2 + 2M$$

M = mutual inductance between L_1 and L_2 .

circuit operation:

When the circuit is turned on, the capacitor is charged. When the capacitor is full charged, it discharged through coils L_1 & L_2 . The output voltage of the amplifier appears across L_1 & feedback voltage across L_2 . The voltage across L_2 is 180° out of phase with the voltage developed across L_1 (V_{out}) as shown in the circuit.



* Electronic Oscillator & its essential components:

1. Amplifier

2. Feedback network

3. Frequency selective network

4. DC biasing circuit

5. Output filter

6. Modulation circuit

7. Frequency divider

8. Frequency multiplier

9. Frequency doubler

10. Frequency doubler

11. Frequency doubler

12. Frequency doubler

13. Frequency doubler

14. Frequency doubler

15. Frequency doubler

16. Frequency doubler

17. Frequency doubler

18. Frequency doubler

19. Frequency doubler

20. Frequency doubler